

BFS(Breadth first Search) And DFS (Depth first Search)

```
from collections import deque
```

```
def bfs(graph, start):
```

```
    visited = set()
```

```
    queue = deque([start])
```

```
    visited.add(start)
```

```
    order = []
```

```
    while queue:
```

```
        vertex = queue.popleft()
```

```
        order.append(vertex)
```

```
        for neighbor in graph[vertex]:
```

```
            if neighbor not in visited:
```

```
                visited.add(neighbor)
```

```
                queue.append(neighbor)
```

```
    return order
```

```
def dfs(graph, start):
```

```
    visited = set()
```

```
    stack = [start] # Python list works perfectly as a LIFO stack
```

```
    order = []
```

```
    while stack:
```

```
        vertex = stack.pop() # .pop() removes and returns the LAST item
```

```
        if vertex not in visited:
```

```
            visited.add(vertex)
```

```
            order.append(vertex)
```

```
        # Reverse neighbors to explore left-to-right (matches BFS ordering logic)
```

```
        for neighbor in reversed(graph[vertex]):
```

```
            if neighbor not in visited:
```

```
                stack.append(neighbor)
```

```
    return order
```

```
# Example usage to test the functions:
```

```
if __name__ == "__main__":
```

```
    example_graph = {
```

```
        'A': ['B', 'C'],
```

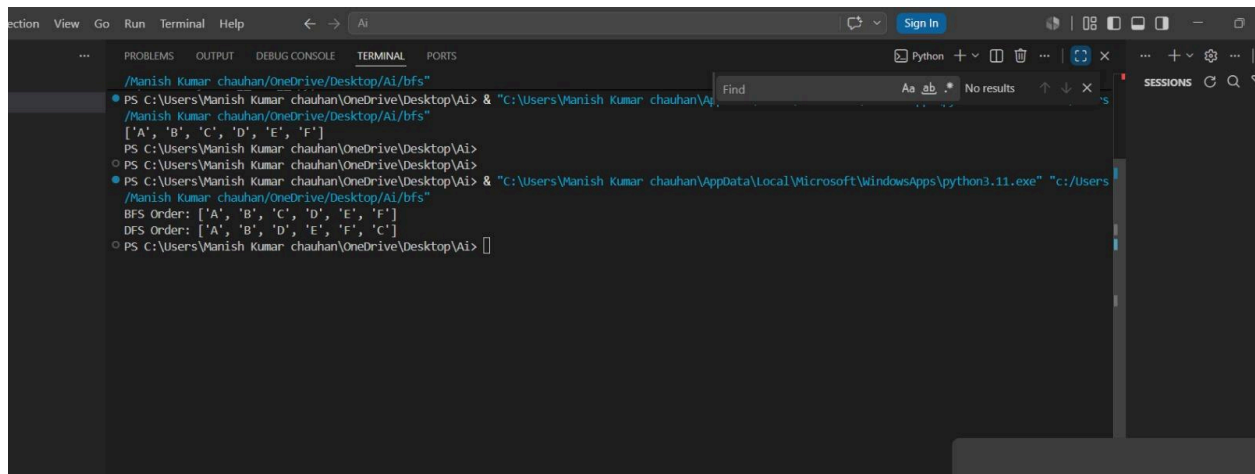
```
        'B': ['A', 'D', 'E'],
```

```
        'C': ['A', 'F'],
```

```
'D': ['B'],
'E': ['B', 'F'],
'F': ['C', 'E']
}

print("BFS Order:", bfs(example_graph, 'A'))
print("DFS Order:", dfs(example_graph, 'A'))
```

OUTPUT:



The screenshot shows a VS Code terminal window with the following content:

```
PS C:\Users\Manish Kumar chauhan\OneDrive\Desktop\Ai> & "C:\Users\Manish Kumar chauhan\AppData\Local\Microsoft\WindowsApps\python3.11.exe" "c:/Users/Manish Kumar chauhan/OneDrive/Desktop/Ai/bfs"
['A', 'B', 'C', 'D', 'E', 'F']
PS C:\Users\Manish Kumar chauhan\OneDrive\Desktop\Ai>
PS C:\Users\Manish Kumar chauhan\OneDrive\Desktop\Ai> & "C:\Users\Manish Kumar chauhan\AppData\Local\Microsoft\WindowsApps\python3.11.exe" "c:/Users/Manish Kumar chauhan/OneDrive/Desktop/Ai/dfs"
BFS Order: ['A', 'B', 'C', 'D', 'E', 'F']
DFS Order: ['A', 'B', 'D', 'E', 'F', 'C']
PS C:\Users\Manish Kumar chauhan\OneDrive\Desktop\Ai>
```